Compatibility of Local Descriptions as a Reconstruction Principle for Quantum Theory and Spacetime

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Abstract

We investigate the consequences of a minimal consistency requirement: whenever observers can compare predictions about the same event, their probability assignments must agree. We formalize this compatibility condition for overlapping partial descriptions without assuming quantum states, spacetime geometry, or thermodynamic laws.

We show that compatibility together with contextuality forces a non-Boolean event structure whose continuous reversible symmetries lead to Hilbert-space probability geometry and the Born rule. Requiring agreement to persist under time evolution restricts dynamics to unitary transformations. Imposing finite reconciliation speed between observers determines Lorentzian causal structure. Limiting reconciliation across spatial partitions produces an area law for accessible information, and observer-independent entropy accounting then recovers Einstein gravitational dynamics in the sense of thermodynamic gravity.

The framework presents quantum theory, relativistic causality, and gravitational dynamics as successive constraints required for mutually coherent distributed description, rather than independent postulates.

1 Introduction

Modern theoretical physics combines quantum theory, relativity, and thermodynamics. Traditionally these frameworks are postulated independently and later unified. Here we explore a different possibility: all three may arise from a single structural requirement governing shared descriptions.

Consider multiple observers assigning probabilities to physical events using incomplete information. Whenever observers can later compare predictions about the same event, agreement must occur. We elevate this operational fact to a principle.

> **Compatibility Principle:** If two observers can jointly test a proposition, they must assign it the same probability.

This statement refers only to mutual consistency of partial descriptions. It assumes neither global states nor spacetime geometry.

Each step below should be read as a conditional derivation: given the stated operational assumptions, the corresponding mathematical structure is uniquely selected.

We show that progressively enlarging the domain over which compatibility must hold forces increasingly restrictive mathematical structure:

$$\text{Compatibility} \Rightarrow \text{Quantum Structure} \Rightarrow \text{Causal Structure} \Rightarrow \text{Gravitational Dynamics}$$

The goal is not to determine the detailed laws of a particular universe, but to characterize the minimal structure required for distributed physical descriptions to remain mutually coherent.

The auxiliary assumptions do not introduce independent physical postulates; they specify the domains over which compatibility must remain stable (refinement, time, signaling, partition)

2 Partial Descriptions and Logical Structure

We begin without assuming quantum theory. Observers possess only incomplete information and assign probabilities within limited measurement domains.

2.1 Observers and Contexts

Definition 1 (Measurement Context)**.** *A context C is a maximal set of mutually co-testable propositions.*

Definition 2 (Partial Description)**.** *An observer assigns a probability function*

$$P_C : C \to [0, 1]$$

satisfying additivity on mutually exclusive outcomes.

Definition 3 (Compatibility)**.** *Two observers with contexts C_1, C_2 are compatible if*

$$P_{C_1}(A) = P_{C_2}(A), \quad \forall A \in C_1 \cap C_2.$$

No global state is assumed. Only local consistency across overlaps is required.

2.2 Operational Assumptions

We formalize properties implicit in probabilistic experiments.

(L1) **Exclusive Additivity:** If $A \perp B$ then

$$P(A \vee B) = P(A) + P(B).$$

(L2) **Complementarity:** For every A there exists $A^{\perp}$ such that

$$P(A) + P(A^{\perp}) = 1.$$

(L3) **Refinement Stability:** If $A \leq B$, performing the finer measurement B cannot change previously agreed probability for A.

(L4) **Contextuality:** No global valuation assigning definite values to all propositions exists.

2.3 Lattice Structure

From (L1–L2) propositions form an orthocomplemented lattice with operations $A \wedge B$ and $A \vee B$. Classical logic would further require distributivity:

$$A \wedge (B \vee C) = (A \wedge B) \vee (A \wedge C).$$

However contextuality forbids global truth assignments.

Proposition 1 (Non-Booleanity)**.** *Compatibility together with contextuality implies the event algebra is non-distributive.*

Proof. If distributivity held, the lattice would be Boolean. Boolean algebras admit two-valued homomorphisms generating all probability measures as convex combinations of deterministic assignments. This contradicts contextuality assumption (L4). Therefore the algebra cannot be Boolean. □

2.4 Orthomodularity

We now determine the weakest logical structure consistent with compatibility.

We impose Mackey consistency: probabilities assigned to a coarse event must equal the sum of probabilities of any measurable refinement.

Theorem 1 (Orthomodularity from Compatibility)**.** *The lattice of propositions satisfies the orthomodular law*

$$A \leq B \;\Rightarrow\; B = A \vee (A^{\perp} \wedge B).$$

Proof. Let AB and suppose the orthomodular identity fails: $BA(AB)$.

Then there exist two distinct decompositions of B into mutually exclusive components, one containing A and one not equivalent to it.

By refinement stability (L3), the probability assigned to A must equal the sum of probabilities of any refinement components projecting onto A.

Different decompositions therefore give different probabilities for A, contradicting compatibility across contexts.

Hence the orthomodular identity must hold. □

Corollary 1. *The orthomodular lattice is the weakest event structure consistent with contextuality and compatibility.*

Stronger structures become Boolean and contradict contextuality, while weaker ones allow refinement-dependent probabilities and violate compatibility.

3 From Logical Structure to Hilbert Space Geometry

Having established that propositions form an orthomodular lattice, we now determine the structure of states compatible with continuous comparison of outcomes.

3.1 Operational Assumptions

We add assumptions required for reproducible experiments:

(G1) **Continuous Reversible Transformations:** Minimal outcomes can be continuously transformed into each other.

(G2) **Observer-Independent Transition Probabilities:** For minimal propositions a, b, the transition probability $p(a, b)$ is the same for all observers.

(G3) **Additivity over Exclusive Refinements:** If $b_1 \perp b_2$ then

$$p(a, b_1 \vee b_2) = p(a, b_1) + p(a, b_2).$$

(G4) **Arbitrary Refinement:** Measurements admit arbitrarily fine decompositions.

Minimal propositions will be called *atoms*.

3.2 Transition Probabilities

For atoms a, b, define $p(a, b)$ as the probability that outcome b occurs when the system is prepared in a.

Properties:

$$0 \leq p(a, b) \leq 1, \qquad p(a, a) = 1, \qquad p(a, b) = p(b, a).$$

Within a two-outcome context $\{b, b^{\perp}\}$,

$$p(a, b) + p(a, b^{\perp}) = 1.$$

Thus pairs of atoms behave like points on a unit circle parameterized by a parameter θ_{ab}.

3.3 Composition Constraint

Compatibility across intermediate comparisons requires consistency:

If a is compared to c via b, the result cannot depend on the chosen intermediate refinement. Hence there exists a function F such that

$$p(a, c) = F(p(a, b), p(b, c)).$$

Symmetry and continuity restrict admissible forms of F.

3.4 Quadratic Form

Theorem 2 (Hilbert-compatible transition law)**.** *The only continuous probability function compatible with symmetry, additivity, and reversible transformations is*

$$p(a, b) = \cos^2(\theta_{ab}).$$

Proof. Continuous reversible transformations act transitively on atoms, so transition probabilities depend only on a single invariant parameter θ_{ab}.

Additivity over orthogonal decompositions implies

$$p(a, b) + p(a, b^{\perp}) = 1.$$

Associativity of intermediate comparisons requires a composition rule invariant under rotations of contexts. The only continuous functions satisfying these constraints are quadratic in a cosine parameter. Linear dependence would violate additivity under refinement, and higher powers violate normalization under composition.

These constraints uniquely characterize a projective inner-product geometry. For orthomodular lattices admitting continuous reversible symmetries and arbitrary refinements, the representation is projective Hilbert space and transition probabilities take the quadratic Born form p(a,b)=—⟨a,b⟩—², parameterizable as cos². □

3.5 Inner Product

Define a complex amplitude

$$\langle a, b \rangle = e^{i\phi} \cos(\theta_{ab}).$$

Then

$$p(a, b) = |\langle a, b \rangle|^2.$$

Orthogonality corresponds to zero overlap. Additivity defines linear span of atoms.

Proposition 2. *Atoms correspond to one-dimensional subspaces of a vector space endowed with an inner product.*

Completeness under arbitrary refinement yields a Hilbert space.

Corollary 2. *Compatibility determines a Hilbert space probability geometry and the Born rule.*

4 Dynamics from Preservation of Compatibility

We now extend compatibility from static comparisons to temporal evolution.

4.1 Temporal Assumptions

(D1) Observers who agree at one time must agree after evolution when comparing the same proposition.

(D2) Time evolution is continuous.

(D3) Time evolution is reversible for isolated systems.

Let T denote the evolution map acting on states.

4.2 Transition Probability Preservation

Agreement on all measurable outcomes requires that transition probabilities remain invariant under evolution.

Proposition 3 (Probability Preservation)**.** *For all states ψ, ϕ,*

$$p(\psi, \phi) = p(T\psi, T\phi).$$

Proof. Two observers preparing ψ and ϕ must later agree on measurement statistics. If transition probabilities changed, observers related by time evolution would assign different probabilities to the same outcome, violating compatibility. □

4.3 Symmetry of Transition Probabilities

Theorem 3. *Any transformation preserving transition probabilities is implemented by a unitary or antiunitary operator.*

Proof. In a Hilbert geometry, mappings preserving squared inner products correspond to projective symmetries. Such maps lift to unitary or antiunitary operators. □

4.4 Continuity

Proposition 4. *Continuous reversible evolution excludes antiunitary transformations.*

Proof. Antiunitary operators involve complex conjugation and cannot form continuous one-parameter groups. Since time evolution is continuous (D2), only unitary operators remain admissible. □

4.5 Generator of Evolution

Theorem 4 (Unitary Dynamics)**.** *There exists a self-adjoint operator H such that*

$$U(t) = e^{-iHt/\hbar}.$$

Proof. A strongly continuous unitary one-parameter group admits a self-adjoint generator by the generator theorem for strongly continuous unitary groups. □

Corollary 3 (Schrödinger Equation)**.** *States satisfy*

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = H |\psi(t)\rangle .$$

4.6 Interpretation

Unitary dynamics is therefore not an independent postulate. It is the unique evolution law preserving compatibility of probability assignments over time.

5 Causal Structure from Finite Reconciliation

We now analyze compatibility between spatially separated observers.

5.1 Operational Assumptions

(C1) Observers compare records only via physical signals.

(C2) Signals propagate with a finite maximal speed c.

(C3) Observers can synchronize clocks using two-way exchange.

(C4) No preferred observer exists.

5.2 Causal Order

Define the relation

$$A \prec B$$

if a signal emitted at event A can reach event B.

Finite signal speed partitions events into reachable and unreachable sets.

Proposition 5. *Events form a partially ordered set with a causal cone boundary satisfying*

$$|\Delta x| \leq c|\Delta t|.$$

5.3 Operational Interval

Observers determine separations using round-trip signaling:

send at t_1, reflect at t_2, receive at t_3.

Define

$$\Delta t = \frac{t_3 - t_1}{2}, \qquad \Delta x = \frac{c(t_3 - t_2)}{2}.$$

These quantities depend only on mutual exchange and must be observer-independent.

5.4 Allowed Transformations

Transformations between observers must:

- preserve the causal boundary,
- preserve round-trip results,
- form a continuous group.

5.5 Lorentz Invariance

Theorem 5. *The only continuous transformations preserving causal ordering and round-trip signaling preserve the quadratic interval*

$$ds^2 = -c^2dt^2 + dx^2 + dy^2 + dz^2.$$

Proof. Galilean transformations preserve absolute simultaneity and change round-trip signaling results, violating (C3). Direction-dependent metrics change two-way signal measurements and violate mutual synchronization. The remaining continuous transformations preserving a universal maximal signal speed and reciprocity form the Lorentz group. □

Corollary 4. *Coordinates between observers are related by*

$$x' = \gamma(x - vt), \qquad t' = \gamma\left(t - \frac{vx}{c^2}\right).$$

5.6 Interpretation

Lorentz spacetime is the minimal structure ensuring that observers never disagree about which events could have influenced which others after exchanging signals.

6 Boundary Information and Gravitational Dynamics

We now consider observers restricted to spatial regions and require compatibility of descriptions across partitions.

6.1 Operational Assumptions

(T1) Observers access only finite regions of space.

(T2) Information exchange occurs only through boundaries of regions.

(T3) Signals carry finite distinguishable information per unit time.

(T4) Entropy measures the additional information required to reconcile two descriptions.

6.2 Localization of Disagreement

Consider observers inside and outside a region A.

Any difference between their descriptions must have crossed the boundary ∂A. Interior processes that never affect the boundary cannot influence mutual agreement.

Proposition 6. *All reconcilable differences between observers in complementary regions are mediated by boundary crossings.*

6.3 Channel Counting

During time Δt, signals cross only a boundary layer of thickness $c\Delta t$.

Independent signals must occupy distinguishable boundary patches. Hence the number of independent communication channels satisfies

$$N \propto \text{Area}(\partial A).$$

6.4 Area Law

Theorem 6 (Boundary Information Bound)**.** *The maximal entropy compatible with mutual reconciliation satisfies*

$$S(A) \leq \eta \, Area(\partial A).$$

Proof. Entropy equals the minimal additional information required to reconcile descriptions. Since information transfer occurs only through independent boundary channels and their number scales with boundary area, the maximal entropy is proportional to the area rather than the volume. □

6.5 Energy Flow

Energy crossing the boundary changes the number of distinguishable states accessible to observers.

Let δQ be the energy flux across a local boundary and δS the corresponding change in reconciliation entropy.

Compatibility requires observers to agree on the state count, implying a universal conversion rule

$$\delta Q = T \, \delta S.$$

Observer-independent entropy accounting takes the Clausius form, whose local implementation yields Einstein dynamics.

6.6 Geometric Consistency

Demanding the above relation for arbitrarily small local boundaries constrains spacetime curvature.

Theorem 7 (Einstein Dynamics)**.** *Requiring observer-independent entropy balance for all local horizons yields*

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}.$$

Proof. Applying the entropy-area relation together with local energy flux across causal horizons produces a relation between curvature and energy-momentum tensor. The only tensor equation consistent for all observers is the Einstein field equation. □

6.7 Interpretation

Gravity appears as the condition required so that observers never disagree about entropy flow across boundaries.

7 Scope and Limits

The compatibility principle constrains the mathematical form of admissible physical theories but does not determine detailed dynamics.

The framework derives structural features common to all compatible theories:

- non-Boolean logical structure,
- Hilbert space probability geometry,
- unitary time evolution,
- Lorentz causal structure,
- boundary area information scaling,
- Einstein gravitational dynamics.

However, it does not determine contingent physical content, including:

- particle spectrum,
- gauge group,
- coupling constants,
- cosmological constant,
- spacetime dimensionality,
- initial conditions.

These correspond to specific realizations within the compatible class rather than logical necessities.

8 Minimality of the Constraint Chain

At each stage the admissible theory space is reduced by requiring compatibility across a larger domain of comparison:

Domain of agreement	Forced structure
overlapping measurements	orthomodular logic
context transformations	Hilbert space geometry
temporal evolution	unitary dynamics
signal exchange	Lorentz causality
spatial partitions	area information bound
local horizons	Einstein gravity

Alternative structures either contradict contextuality, permit disagreement between observers, or violate finite reconciliation constraints.

The argument should be understood as a chain of necessary conditions. Each enlargement of the domain over which observers must remain compatible eliminates classes of theories. Known physical theories lie in the intersection of these constraints.

9 Conclusion

We have investigated a single operational requirement: observers assigning probabilities to shared events must agree whenever comparison is possible.

Without assuming quantum states, spacetime geometry, or thermodynamic laws, this requirement progressively restricts admissible physical descriptions.

The resulting structure is:

$$\text{Compatibility} \Rightarrow \text{Quantum Theory} \Rightarrow \text{Relativistic Causality} \Rightarrow \text{Gravitational Dynamics}.$$

Quantum theory governs consistent probability assignments, relativity governs consistent comparison of events, and gravity governs consistent entropy accounting across horizons.

Rather than independent principles requiring unification, these frameworks appear as different aspects of the same consistency condition applied across measurement, time, space, and partition.

10 Outlook

The present work determines structural constraints but not detailed dynamics. Possible extensions include deriving gauge structure from patch compatibility and constraining effective field theories from reconciliation networks.